

Exercises - 3

1. (a) Given that,

$$A = \begin{bmatrix} 1 & 0 & 3 & 4 \\ 5 & 2 & 0 & 1 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 6 & -1 & -2 & 5 \\ 1 & 0 & -3 & 4 \end{bmatrix}$$

$$I = A^T A = T A A$$

$$\therefore 3A = 3 \times \begin{bmatrix} 1 & 0 & 3 & 4 \\ 5 & 2 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 0 & 9 & 12 \\ 15 & 6 & 0 & 3 \end{bmatrix}$$

$$A+B = \begin{bmatrix} 1 & 0 & 3 & 4 \\ 5 & 2 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 6 & -1 & -2 & 5 \\ 1 & 0 & -3 & 4 \end{bmatrix} = \begin{bmatrix} 7 & -1 & 1 & 9 \\ 6 & 2 & -3 & 5 \end{bmatrix}$$

$$A-B = \begin{bmatrix} 1 & 0 & 3 & 4 \\ 5 & 2 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 6 & -1 & -2 & 5 \\ 1 & 0 & -3 & 4 \end{bmatrix} = \begin{bmatrix} -5 & 1 & 5 & -1 \\ 4 & 2 & 3 & -3 \end{bmatrix}$$

$$3A-2B = \begin{bmatrix} 3 & 0 & 9 & 12 \\ 15 & 6 & 0 & 3 \end{bmatrix} - 2 \times \begin{bmatrix} 6 & -1 & -2 & 5 \\ 1 & 0 & -3 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 3 & 0 & 9 & 12 \\ 15 & 6 & 0 & 3 \end{bmatrix} - \begin{bmatrix} 12 & -2 & -4 & 10 \\ 2 & 0 & -6 & 8 \end{bmatrix}$$

$$= \begin{bmatrix} -9 & 2 & 13 & 2 \\ 13 & 6 & 6 & -5 \end{bmatrix}$$

$$(b) A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$\therefore A^2 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \times \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

$$B = \frac{1}{2} \times \begin{bmatrix} -1 & \sqrt{3} \\ \sqrt{3} & 1 \end{bmatrix} = \begin{bmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix}$$

$$\therefore B^2 = \begin{bmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix} \times \begin{bmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix} = \begin{bmatrix} \frac{1}{4} + \frac{3}{4} & -\frac{\sqrt{3}}{4} + \frac{\sqrt{3}}{4} \\ -\frac{\sqrt{3}}{4} + \frac{\sqrt{3}}{4} & \frac{3}{4} + \frac{1}{4} \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

$$C = \frac{1}{2} \times \begin{bmatrix} -1 & -\sqrt{3} \\ -\sqrt{3} & 1 \end{bmatrix} = \begin{bmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix}$$

$$\therefore C^2 = \begin{bmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix} \times \begin{bmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix} = \begin{bmatrix} \frac{1}{4} + \frac{3}{4} & \frac{\sqrt{3}}{4} - \frac{\sqrt{3}}{4} \\ \frac{\sqrt{3}}{4} - \frac{\sqrt{3}}{4} & \frac{3}{4} + \frac{1}{4} \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

$$AB = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \times \begin{bmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix} = \begin{bmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & -\frac{1}{2} \end{bmatrix} = \frac{1}{2} \begin{bmatrix} -1 & \sqrt{3} \\ -\sqrt{3} & -1 \end{bmatrix} = D$$

$$BA = \begin{bmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix} \times \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{bmatrix} = \frac{1}{2} \begin{bmatrix} -1 & -\sqrt{3} \\ \sqrt{3} & -1 \end{bmatrix} = E$$

3. (b) Given that,

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

$$B = \begin{bmatrix} 0 & 1 & 1 & 0 \\ -1 & 0 & 0 & 1 \\ -1 & 0 & 0 & 1 \\ 0 & -1 & -1 & 0 \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$$

$$AB = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix} \times \begin{bmatrix} 0 & 1 & 1 & 0 \\ -1 & 0 & 0 & 1 \\ -1 & 0 & 0 & 1 \\ 0 & -1 & -1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0-1-1+0 & 0+0+0+0 & 0+0+0+0 & 0+1+1+0 \\ 0+0+0+0 & 1+0+0+1 & 1+0+0-1 & 0+0+0+0 \\ 0+0+0+0 & 1+0+0-1 & 1+0+0-1 & 0+0+0+0 \\ 0-1-1+0 & 0+0+0+0 & 0+0+0+0 & 0+1+1+0 \end{bmatrix}$$

$$= \begin{bmatrix} -2 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -2 & 0 & 0 & 2 \end{bmatrix}$$

$$BA = \begin{bmatrix} 0 & 1 & 1 & 0 \\ -1 & 0 & 0 & 1 \\ -1 & 0 & 0 & 1 \\ 0 & -1 & -1 & 0 \end{bmatrix} \times \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0+1+1+0 & 0+0+0+0 & 0 & 0+1+1+0 \\ 0 & -1+0+0+1 & -1+0+0+1 & 0 \\ 0 & -1+0+0+1 & -1+0+0+1 & 0 \\ 0-1-1+0 & 0 & 0 & 0-1-1+0 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -2 & 0 & 0 & -2 \end{bmatrix}$$

$$AB - BA = \begin{bmatrix} -2 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -2 & 0 & 0 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} -4 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix}$$

$$= -4 \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$$

$$= -4C \quad [\text{shown}]$$

$$7. \quad A = \begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{bmatrix}$$

$$\therefore A^2 = \begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{bmatrix} \times \begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{bmatrix}$$

$$= \begin{bmatrix} 16+6-6 & 24+18-24 & 24+12-18 \\ 4+3-2 & 6+9-8 & 6+6-6 \\ -4-4+3 & -6-12+12 & -6-8+9 \end{bmatrix}$$

$$= \begin{bmatrix} 16 & 18 & 18 \\ 5 & 7 & 6 \\ -5 & -6 & -5 \end{bmatrix}$$

$$\text{Now, } A^3 = \begin{bmatrix} 16 & 18 & 18 \\ 5 & 7 & 6 \\ -5 & -6 & -5 \end{bmatrix} \times \begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{bmatrix}$$

$$= \begin{bmatrix} 64 & 78 & 78 \\ 21 & 27 & 26 \\ -21 & -28 & -27 \end{bmatrix}$$

∴ L.H.S.

$$A^3 - 4A^2 - A + 4I = 0$$

$$= \begin{bmatrix} 64 & 78 & 78 \\ 21 & 27 & 26 \\ -21 & -28 & -27 \end{bmatrix} - \begin{bmatrix} 64 & 72 & 72 \\ 20 & 28 & 24 \\ -20 & -24 & -20 \end{bmatrix} - \begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{bmatrix} + \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$= 0$$

[proved]

9. $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 3 \\ 2 & 4 & 9 \end{bmatrix}$

∴ $A^T = \begin{bmatrix} 1 & 2 & 2 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{bmatrix}$

$B = \begin{bmatrix} 0 & -1 & -1 \\ -3 & 2 & 4 \\ 1 & 1 & 0 \end{bmatrix}$

∴ $B^T = \begin{bmatrix} 0 & -3 & 1 \\ 1 & 2 & 1 \\ -1 & 4 & 0 \end{bmatrix}$

$$(AB)^T = \left\{ \begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 3 \\ 2 & 4 & 9 \end{bmatrix} \times \begin{bmatrix} 0 & 1 & -1 \\ -3 & 2 & 4 \\ 1 & 1 & 0 \end{bmatrix} \right\}^T = \begin{bmatrix} -2 & 4 & 3 \\ -3 & 9 & 6 \\ -3 & 10 & 14 \end{bmatrix}^T = \begin{bmatrix} -2 & -3 & -3 \\ 4 & 9 & 19 \\ 3 & 6 & 14 \end{bmatrix}$$

$$B^T A^T = \begin{bmatrix} 0 & -3 & 1 \\ 1 & 2 & 1 \\ -1 & 4 & 0 \end{bmatrix} \times \begin{bmatrix} 1 & 2 & 2 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{bmatrix} = \begin{bmatrix} -2 & -3 & -3 \\ 4 & 9 & 19 \\ 3 & 6 & 14 \end{bmatrix}$$

∴ $(AB)^T = B^T A^T$ [proved]

20] $A = \begin{bmatrix} 1 & -1 & 1 \\ 2 & -1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$

$$0 = I A + A - I A - A$$

$$\therefore A^2 = \begin{bmatrix} 1 & -1 & 1 \\ 2 & -1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \times \begin{bmatrix} 1 & -1 & 1 \\ 2 & -1 & 0 \\ 1 & 0 & 0 \end{bmatrix} =$$

$$= \begin{bmatrix} 0 & 0 & 1 \\ 0 & -1 & 2 \\ 1 & -1 & 1 \end{bmatrix}$$

$$A^3 = A \cdot A^2 = \begin{bmatrix} 1 & -1 & 1 \\ 2 & -1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \times \begin{bmatrix} 0 & 0 & 1 \\ 0 & -1 & 2 \\ 1 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$$

Again, $A^3 = A^2 \cdot A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & -1 & 2 \\ 1 & -1 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & -1 & 1 \\ 2 & -1 & 0 \\ 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$

$$\therefore A^3 = A^2 \cdot A = A \cdot A^2 = I$$

[Showed]

Now,

$$A^3 = I$$

$$\Rightarrow I = A^3$$

multiplying both side with A^{-1}

$$\Rightarrow A^{-1} I = A^3 \cdot A^{-1}$$

$$\Rightarrow \therefore A^{-1} = A^2 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & -1 & 2 \\ 1 & -1 & 1 \end{bmatrix}$$

22) (i) $A = \begin{bmatrix} 2 & -1 & -1 \\ 1 & -2 & 1 \\ 1 & -1 & 2 \end{bmatrix}$

$$\therefore D = |A| = 2(-4+1) + (2-1) - 1(-1+2) = -6 + 1 - 1 = -6 \neq 0$$

Thus A has an inverse matrix. Now,

$$A_{11} = -3, \quad A_{12} = -1, \quad A_{13} = 1$$

$$A_{21} = 3, \quad A_{22} = 5, \quad A_{23} = 1$$

$$A_{31} = -3, \quad A_{32} = -3, \quad A_{33} = -3$$

So, $\text{Adj } A = \begin{bmatrix} -3 & 3 & -3 \\ -1 & 5 & -3 \\ 1 & 1 & -3 \end{bmatrix}$

$$\begin{aligned} \therefore A^{-1} &= \frac{1}{D} \times \text{Adj } A \\ &= -\frac{1}{6} \begin{bmatrix} -3 & 3 & -3 \\ -1 & 5 & -3 \\ 1 & 1 & -3 \end{bmatrix} \\ &= \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{6} & -\frac{5}{6} & \frac{1}{2} \\ -\frac{1}{6} & -\frac{1}{6} & \frac{1}{2} \end{bmatrix} \end{aligned}$$

Ans

$$\underline{16)} \quad A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 3 & 5 \\ 1 & 5 & 12 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 1 & -1 \\ 1 & 2 & 3 \\ 1 & 4 & 9 \end{bmatrix}$$

$$D_1 = |A| = (36 - 25) - 2(12 - 5) + 3(5 - 3)$$

$$= 11 - 14 + 6 = 3 \neq 0$$

$$D_2 = |B| = 6 - 6 + 2 = 2 \neq 0$$

Thus A and B has inverse matrix.

Now.

$$A_{11} = 11$$

$$A_{12} = -7$$

$$A_{13} = 2$$

$$A_{21} = -9$$

$$A_{22} = 9$$

$$A_{23} = -3$$

$$A_{31} = 1$$

$$A_{32} = -2$$

$$A_{33} = 1$$

$$\text{So, } \text{Adj } A = \begin{bmatrix} 11 & -9 & 1 \\ -7 & 9 & -2 \\ 2 & -3 & 1 \end{bmatrix} \therefore A^{-1} = \frac{1}{3} \begin{bmatrix} 11 & -9 & 1 \\ -7 & 9 & -2 \\ 2 & -3 & 1 \end{bmatrix}$$

Again,

$$B_{11} = 6$$

$$B_{12} = -6$$

$$B_{13} = 2$$

$$B_{21} = -5$$

$$B_{22} = 8$$

$$B_{23} = -3$$

$$B_{31} = 1$$

$$B_{32} = -2$$

$$B_{33} = 1$$

$$\text{So, } \text{Adj } B = \begin{bmatrix} 6 & -5 & 1 \\ -6 & 8 & -2 \\ 2 & -3 & 1 \end{bmatrix}$$

$$\therefore B^{-1} = \frac{1}{2} \begin{bmatrix} 6 & -5 & 1 \\ -6 & 8 & -2 \\ 2 & -3 & 1 \end{bmatrix}$$

$$\therefore A^{-1}B = \begin{bmatrix} 11/3 & -3 & 1/3 \\ -7/3 & 3 & -2/3 \\ 2/3 & -1 & 1/3 \end{bmatrix} \times \begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 3 \\ 1 & 1 & 2 \end{bmatrix} \quad (15)$$

$$= \begin{bmatrix} 1 & -1 & -7/3 \\ 0 & 1 & 2/3 \\ 0 & 0 & 2/3 \end{bmatrix}$$

Again, $AB^{-1} = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 3 & 5 \\ 1 & 5 & 12 \end{bmatrix} \times \begin{bmatrix} 2 & -5/2 & 1/2 \\ -3 & 4 & -1 \\ -1 & -3/2 & 1/2 \end{bmatrix}$

$$= \begin{bmatrix} 0 & 1 & 0 \\ -1 & 2 & 0 \\ 0 & -1/2 & 3/2 \end{bmatrix} = L \cdot A \cdot U$$

$$17) A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{bmatrix} \quad \text{adj } A$$

$$D = |A| = 1 - 0 + 0 = 1 \neq 0$$

Thus A has an inverse matrix.
Now,

$$A_{11} = 1$$

$$A_{12} = -1$$

$$A_{13} = -1$$

$$A_{21} = 0$$

$$A_{22} = 1$$

$$A_{23} = 0$$

$$A_{31} = 0$$

$$A_{32} = 0$$

$$A_{33} = 1$$

$$\therefore \text{Adj } A = \begin{bmatrix} 1 & -1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\text{and } A^{-1} = \begin{bmatrix} 1 & -1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\text{Now, } \frac{1}{2} (A^{-1} + A) = \frac{1}{2} \left\{ \begin{bmatrix} 1 & -1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \right\}$$

$$= \frac{1}{2} \times \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$$

$$18. \quad A = \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix} \times \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 3 & -4 & 4 \\ 0 & -1 & 0 \\ -2 & 2 & -3 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 3 & -4 & 4 \\ 0 & -1 & 0 \\ -2 & 2 & -3 \end{bmatrix} \times \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ -2 & 3 & -4 \\ -2 & 3 & -3 \end{bmatrix}$$

Again $D = |A| = 3(-3+4) + 3(2-0) + 4(-2+0)$

$$= 3+6-8 = 1 \neq 0$$

Thus, A has an inverse matrix.

$$A_{11} = 1 \quad A_{12} = -2 \quad A_{13} = -2$$

$$A_{21} = -1 \quad A_{22} = 3 \quad A_{23} = 3$$

$$A_{31} = 0 \quad A_{32} = -4 \quad A_{33} = -3$$

$$\text{So, } \text{Adj} A = \begin{bmatrix} 1 & -1 & 0 \\ -2 & 3 & -4 \\ -2 & 3 & -3 \end{bmatrix}$$

$$\therefore A^{-1} = \begin{bmatrix} 1 & -1 & 0 \\ -2 & 3 & -4 \\ -2 & 3 & -3 \end{bmatrix} = A^3$$

[proved]

26. (ii) Let, $A = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$$A^T = \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A \cdot A^T = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \cos^2\theta + \sin^2\theta & 0 & 0 \\ 0 & \sin^2\theta + \cos^2\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$$

$$A^T \cdot A = \begin{bmatrix} \cos\theta + \sin\theta & 0 & 0 \\ 0 & \cos\theta + \sin\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$$

$$A^T \cdot A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$$

Similarly, $A^T \cdot A = I$

Thus A is an orthogonal matrix.

[Proved]

30. $A = \begin{bmatrix} 2 & -1 & 1 \\ -2 & 3 & -2 \\ -4 & 4 & -3 \end{bmatrix}$

$\lambda = 5 + 6 + 25$ (ii) $\therefore$

$\lambda = 5 + 6 - 2$

$\lambda = 5 - 6 + 25$

$\therefore A^2 = \begin{bmatrix} 2 & -1 & 1 \\ -2 & 3 & -2 \\ -4 & 4 & -3 \end{bmatrix}$

$\times \begin{bmatrix} 2 & -1 & 1 \\ -2 & 3 & -2 \\ -4 & 4 & -3 \end{bmatrix}$

row 1 $\rightarrow$ row 1

row 2 $\rightarrow$ row 2

0 -

$\theta^T A = \chi \theta$

$\begin{bmatrix} 2 \\ -2 \\ -4 \end{bmatrix} = \begin{bmatrix} 2 & -1 & 1 \\ -2 & 3 & -2 \\ -4 & 4 & -3 \end{bmatrix} \begin{bmatrix} 5 \\ 6 \\ 25 \end{bmatrix}$

$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} = A$

row 1

$= A$

$0 \neq \lambda = 5 + (5-1) - 0 = |A| = 0$

Thus A is idempotent : [showed]

$\lambda = 5 \& A$

$\lambda = 5 \& A$

$\lambda = 1 \& A$

$\lambda = 5 \& A$

$\lambda = 5 \& A$

$\lambda = 1 \& A$

$\begin{bmatrix} 5 & 5 & 0 \\ 0 & 5 & 0 \\ 5 & 5 & 5 \end{bmatrix}$

$= A^2 \& A$

$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

$= I - A$

35. (ii) $x + y + z = 6$

$x - y + z = 2$

$2x + y - z = 1$

We can form

write this system into the form $AX = B$

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 2 & 1 & -1 \end{bmatrix} X = \begin{bmatrix} 6 \\ 2 \\ 1 \end{bmatrix}$$

the system into the form $AX = B$

$\therefore X = A^{-1}B$ — (1)

where,

$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 2 & 1 & -1 \end{bmatrix}$

$X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, B = \begin{bmatrix} 6 \\ 2 \\ 1 \end{bmatrix}$

Now,

$D = |A| = 0 - 1(-3) + 3 = 6 \neq 0$

$A_{11} = 0$

$A_{12} = 3$

$A_{13} = 3$

$A_{21} = 2$

$A_{22} = -3$

$A_{23} = 1$

$A_{31} = 2$

$A_{32} = 0$

$A_{33} = -2$

So, $\text{Adj } A = \begin{bmatrix} 0 & 2 & 2 \\ 3 & -3 & 0 \\ 3 & 1 & -2 \end{bmatrix}$

$\therefore A^{-1} = \begin{bmatrix} 0 & 1/3 & 1/3 \\ 1/2 & -1/2 & 0 \\ 1/2 & 1/6 & -1/3 \end{bmatrix}$

Now, $(8-F)A = (2+5) \cdot 2 + (5+4) \cdot 1 = 15$

$$X = A^{-1}B = \begin{bmatrix} 0 & 1/3 & 1/3 \\ 1/2 & -1/2 & 0 \\ 1/2 & 1/6 & -1/3 \end{bmatrix} \begin{bmatrix} 6 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$$\therefore \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

Hence $x = 1, y = 2, z = 3.$

36. (i) $5x - 6y + 4z = 15$

$7x + 4y - 3z = 19$

$2x + y + 6z = 46$

We can write this system into the form

$$AX = B$$

$$\therefore X = A^{-1}B$$

where,

$$A = \begin{bmatrix} 5 & -6 & 4 \\ 7 & 4 & -3 \\ 2 & 1 & 6 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, B = \begin{bmatrix} 15 \\ 19 \\ 46 \end{bmatrix}$$

Now, $D = |A| = 5(24+3) + 6(42+6) + 4(7-8)$
 $= 135 + 288 - 4$
 $= 419 \neq 0$

$$A_{11} = 27$$

$$A_{12} = -48$$

$$A_{13} = -1$$

$$A_{21} = 46$$

$$A_{22} = 22$$

$$A_{23} = -17$$

$$A_{31} = 2$$

$$A_{32} = 43$$

$$A_{33} = 62$$

So, $\text{Adj } A = \begin{bmatrix} 27 & 40 & 2 \\ -48 & 22 & 43 \\ -1 & -17 & 62 \end{bmatrix}$

$$\therefore A^{-1} = \frac{1}{419} \begin{bmatrix} 27 & 40 & 2 \\ -48 & 22 & 43 \\ -1 & -17 & 62 \end{bmatrix}$$

Again,

$$X = A^{-1}B = \frac{1}{419} \begin{bmatrix} 27 & 40 & 2 \\ -48 & 22 & 43 \\ -1 & -17 & 62 \end{bmatrix} \times \begin{bmatrix} 115 \\ 19 \\ 46 \end{bmatrix}$$

$$= \frac{1}{419} \begin{bmatrix} 1257 \\ 1676 \\ 2574 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \\ 6 \end{bmatrix}$$

$$\therefore \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \\ 6 \end{bmatrix}$$

Hence $x = 3, y = 4, z = 6$

$$(ii) \quad 2x + 3y + 4z = 1$$

$$3x + 4y - 5z = 10$$

$$5x - 7y + 2z = 3$$

We can write this system into the form,

$$AX = B$$

$$\therefore X = A^{-1} B$$

where,

$$A = \begin{bmatrix} 2 & 3 & 4 \\ 3 & 4 & -5 \\ 5 & -7 & 2 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 10 \\ 3 \end{bmatrix}$$

$$\text{Now, } D = |A| = -54 + 93 - 164 = -125 \neq 0$$

$$A_{11} = -27$$

$$A_{12} = -31$$

$$A_{13} = -41$$

$$A_{21} = -22$$

$$A_{22} = -16$$

$$A_{23} = -1$$

$$A_{31} = -1$$

$$A_{32} = 22$$

$$A_{33} = 17$$

$$\text{So, } \text{Adj } A = \begin{bmatrix} -27 & -22 & -1 \\ -31 & -16 & 22 \\ -41 & -1 & 17 \end{bmatrix}$$

$$\therefore A^{-1} = \frac{1}{|A|} \text{Adj } A = \frac{1}{-125} \begin{bmatrix} -27 & -22 & -1 \\ -31 & -16 & 22 \\ -41 & -1 & 17 \end{bmatrix}$$

From (1)

$$X = A^{-1}B = -\frac{1}{125} \begin{bmatrix} -27 & -22 & -1 \\ -31 & -16 & 22 \\ -41 & -1 & 17 \end{bmatrix} X = \begin{bmatrix} 1 \\ 10 \\ 3 \end{bmatrix}$$

$$= -\frac{1}{125} \times \begin{bmatrix} -250 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$$

Hence, $x=2, y=1, z=0$

$$\begin{aligned} \text{(X)} \quad x+2y+3z+4 &= 0 \\ 2x+4y+5z+7 &= 0 \\ 3x+5y+6z+10 &= 0 \end{aligned}$$

we can write this system into

$$X = A^{-1}B$$

$$\text{Where, } A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}, B = \begin{bmatrix} -4 \\ -7 \\ -10 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

Now,

$$D = |A| = -1 + 6 - 6 = -1 \neq 0$$

$$A_{11} = -1, \quad A_{12} = 3, \quad A_{13} = -2$$

$$A_{21} = 3, \quad A_{22} = -3, \quad A_{23} = 1$$

$$A_{31} = -2, \quad A_{32} = 1, \quad A_{33} = 0$$

$$\text{So, Adj } A = \begin{bmatrix} -1 & 3 & 2 \\ 3 & -3 & 1 \\ -2 & 1 & 0 \end{bmatrix}$$

$$\therefore A^{-1} = \frac{1}{|A|} \text{Adj } A = \frac{1}{-10} \begin{bmatrix} -1 & 3 & 2 \\ 3 & -3 & 1 \\ -2 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 0.1 & -0.3 & -0.2 \\ -0.3 & 0.3 & 0.1 \\ 0.2 & -0.1 & 0 \end{bmatrix}$$

$$\text{Now, } X = A^{-1} B = \begin{bmatrix} 0.1 & -0.3 & -0.2 \\ -0.3 & 0.3 & 0.1 \\ 0.2 & -0.1 & 0 \end{bmatrix} \times \begin{bmatrix} -4 \\ -7 \\ -10 \end{bmatrix} = \begin{bmatrix} -3 \\ 1 \\ -1 \end{bmatrix}$$

$$\therefore \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -3 \\ 1 \\ -1 \end{bmatrix}$$

$$\text{Hence, } x = -3, y = 1, z = -1$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$$

$$\begin{aligned}
 35.(v) \quad x + 2y + z &= 2 \\
 2x - y + 2z &= -1 \\
 3x - 4y - 3z &= -16
 \end{aligned}$$

we can write this system into

$$X = A^{-1} B$$

where,

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & -1 & 2 \\ 3 & -4 & -3 \end{bmatrix}, \quad B = \begin{bmatrix} 2 \\ -1 \\ -16 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$\text{Now, } D = |A| = 11 + 24 - 5 = 30 \neq 0$$

$$A_{11} = 11 \quad A_{12} = 12 \quad A_{13} = -5$$

$$A_{21} = 2 \quad A_{22} = -6 \quad A_{23} = 10$$

$$A_{31} = 5 \quad A_{32} = 0 \quad A_{33} = -5$$

$$\text{So } \text{Adj} A = \begin{bmatrix} 11 & 2 & 5 \\ 12 & -6 & 0 \\ -5 & 10 & -5 \end{bmatrix}$$

$$A^{-1} = \frac{1}{30} \begin{bmatrix} 11 & 2 & 5 \\ 12 & -6 & 0 \\ -5 & 10 & -5 \end{bmatrix}$$

P-solve

Again,

$$X = A^{-1}B$$

$$= \frac{1}{30} \begin{bmatrix} 11 & 2 & 5 \\ 12 & -6 & 0 \\ -5 & 10 & -5 \end{bmatrix}$$

$$\times \begin{bmatrix} 2 \\ -1 \\ -16 \end{bmatrix}$$

$$= \frac{1}{30} \begin{bmatrix} -60 \\ 30 \\ 60 \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \\ 2 \end{bmatrix}$$

$$\therefore \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \\ 2 \end{bmatrix}$$

Hence $x = -2$, $y = 1$, $z = 2$

chapter - 4

3. (ii)
$$\begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 5 \\ 2 & 4 & 8 \end{bmatrix}$$

$A^T A = X$

Reduce the matrix into echelon form,

$$r'_2 = -r_1 + r_2$$

$$r'_3 = -2r_1 + r_3$$

$$\sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 2 \\ 0 & 0 & 2 \end{bmatrix}$$

$$r'_3 = -r_2 + r_3$$

$$\sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 2 \end{bmatrix}$$

Hence Rank of this matrix is 2.

$$v) \begin{bmatrix} 0 & 1 & -1 & 3 \\ 1 & 1 & 2 & 3 \\ 1 & -1 & 1 & 4 \end{bmatrix} \begin{bmatrix} 1 & -1 & 1 & 4 \\ 1 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{(1) \leftrightarrow (2)}$$

Reduce this matrix into echelon form and

$$\rightarrow \begin{bmatrix} 1 & -1 & 1 & 4 \\ 1 & 1 & 2 & 3 \\ 0 & 1 & -1 & 3 \end{bmatrix} \begin{bmatrix} 1 & -1 & 1 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{r_2 - r_1}$$

$$r_2' = -r_1 + r_2$$

$$\approx \begin{bmatrix} 1 & -1 & 1 & 4 \\ 0 & 2 & 1 & -1 \\ 0 & 1 & -1 & 3 \end{bmatrix} \begin{bmatrix} 1 & -1 & 1 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{r_2 \leftrightarrow r_3}$$

$$r_3' = -r_2 + 2r_3$$

$$\approx \begin{bmatrix} 1 & -1 & 1 & 4 \\ 0 & 2 & 1 & -1 \\ 0 & 0 & -3 & 7 \end{bmatrix} \begin{bmatrix} 1 & -1 & 1 & 4 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{(1) \leftrightarrow (2)}$$

Hence the rank is 3

$$\begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix} \xrightarrow{r_1 + r_2 - r_3}$$

$$4. (a) A = \begin{bmatrix} 1 & 2 & -1 \\ -1 & 1 & 1 \\ 0 & 3 & 0 \end{bmatrix} \quad \begin{bmatrix} 1 & -1 & 1 & 0 \\ 0 & 5 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

from row 1, $r_2' = r_1 + r_2$

$$\approx \begin{bmatrix} 1 & 2 & -1 \\ 0 & 3 & 0 \\ 0 & 3 & 0 \end{bmatrix} \quad \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 5 & 1 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

$$r_3' = -r_2 + r_3$$

$$\approx \begin{bmatrix} 1 & 2 & -1 \\ 0 & 3 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 5 & 1 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

Hence $\rho(A) = 2$

$$5. (a) A = \begin{bmatrix} 1 & -1 & 2 & 0 \\ 1 & 1 & 1 & 1 \\ -1 & -3 & 0 & -2 \end{bmatrix} \quad \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 5 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$r_2' = -r_1 + r_2, \quad r_3' = r_1 + r_3$$

$$\approx \begin{bmatrix} 1 & -1 & 2 & 0 \\ 0 & 2 & -1 & 1 \\ 0 & -4 & 2 & -2 \end{bmatrix}$$

$$r_3' = 2r_2 + r_3$$

$$\sim \begin{bmatrix} 1 & -1 & 2 & 0 \\ 0 & 2 & -1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Hence $\rho(A) = 2$

$$\sim \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

6. (i) $\begin{bmatrix} -1 & 1 & 3 & 2 \\ 1 & 1 & 0 & -1 \\ 0 & -1 & -1 & 0 \\ 2 & -1 & 1 & 2 \\ 3 & 2 & -2 & -1 \end{bmatrix}$

$$\sim \begin{bmatrix} -1 & 1 & 3 & 2 \\ 1 & 1 & 0 & -1 \\ 0 & -1 & -1 & 0 \\ 2 & -1 & 1 & 2 \\ 3 & 2 & -2 & -1 \end{bmatrix}$$

$$r_2' = r_1 + r_2, \quad r_3' = 3r_1 + r_3$$

$$\sim \begin{bmatrix} -1 & 1 & 3 & 2 \\ 0 & 2 & 3 & 1 \\ 0 & -1 & -1 & 0 \\ 0 & 1 & 7 & 6 \\ 0 & -1 & -1 & 0 \end{bmatrix}$$

$$r_3' = -5r_2 + 2r_3, \quad r_4' = -r_2 + 2r_3, \quad r_5 = r_2 + 2r_3$$

$$\sim \begin{bmatrix} -1 & 1 & 3 & 2 \\ 0 & 2 & 3 & 1 \\ 0 & 0 & -1 & 5 \\ 0 & 0 & 11 & 11 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

$$r_4' = 11r_3 + r_4, \quad r_5' = r_3 + r_5$$

$$\begin{bmatrix} -1 & 1 & 3 & 2 \\ 0 & 2 & 3 & 1 \\ 0 & 0 & -1 & 5 \\ 0 & 0 & 0 & 66 \\ 0 & 0 & 0 & 6 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 5 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$S = (A) \eta$$

correct

$$r_5' = r_4 - 11r_5$$

$$\begin{bmatrix} -1 & 1 & 3 & 2 \\ 0 & 2 & 3 & 1 \\ 0 & 0 & -1 & 5 \\ 0 & 0 & 0 & 66 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 5 \\ 1 & 5 & 5 & 0 \end{bmatrix}$$

$$r_1 + r_2 = r_3, \quad r_1 + r_4 = r_5$$

Hence the rank is

$$\begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

বিকল্পভাবে একইভাবে

$$2r_5 + r_3 = r_4$$

$$r_5 + r_3 = r_4$$

$$2r_5 + r_4 = r_3$$

$$\begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$